
Stock Projections (TOR 6)

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Define and document methods for producing projections; provide justification for assumptions of fishery selectivity, fecundity, mortality and recruitment; comment on the reliability of resulting projections considering the effects of uncertainty and sensitivity to projection assumptions. Compare the results of SAMS and GeoSAMS and comment on their appropriateness for use in management.

Because of the sedentary nature of sea scallops, fishing mortality can vary considerably in space even in the absence of area specific management (Hart 2001). Rotational management and long-term closures exacerbate this heterogeneity. Projections that ignore spatial variation can be unrealistic and misleading. For example, suppose 80% of the stock biomass is in areas closed to fishing (as has occurred in some years in Georges Bank). A stock projection that ignored the closure and assumed an overall F of 0.2 would forecast landings nearly equal to the entire stock biomass in the areas open to fishing. Thus, using a non-spatial forecasting model could lead to unsustainable harvest levels under area management. For these reasons, a spatial forecasting model (the Scallop Area Management Simulator, SAMS) was developed for use in sea scallop management. Various versions of SAMS have been used since 1999. In the current version of SAMS used here, the resource is divided into 24 separate subareas, 7 in the Mid-Atlantic, 12 on Georges Bank, and five in the Gulf of Maine (Figure 4.1).

Population dynamics is modeled similar in SAMS to CASA and SYM, that is, populations are size-structured and use stochastic growth transition matrices. However, processes can be modeled at a finer spatial scale in SAMS compared to the other models. Specifically, each area has its own stochastic growth transition matrix derived from the shell increments collected in that area from the most recent growth period. Similarly, each SAMS area has its own shell height to meat weight relationship, based on data from that area. Natural and fishing mortality and recruitment are also area-specific. Fishing mortality in a subarea can either be explicitly specified in each area, as would be the case for closed areas or “access areas”, which are managed by an area-specific quota, or calculated using a simple fleet dynamics model which assumes fishing effort is proportional to estimated LPUE. Adult M is based on that used for CASA in the most recent period, except in the two southern-most areas in the Mid-Atlantic, Virginia and Delmarva, which are set at 4 and 0.6, respectively.

Projected recruitment is modelled stochastically with the log-transformed mean and covariance for recruitment in each area matching that observed in NEFSC dredge survey

time series. Mean recruitment is then scaled to a region-wide Beverton-Holt stock recruitment relationship, making the SAMS model more comparable to the SYM reference points model. Thus, the recruitment is multiplied by $s_r(B_r)/s_r(B_{\text{mean}})$, where s_r is the (mean) regional stock recruit relationship (for two years old) used in the SYM model, B_r is the current regional biomass, and B_{mean} is the mean overall biomass.

Initial conditions are based on surveys, which are bootstrapped parametrically. The model is run 1000 times, with different initial conditions, and recruitment. Natural mortality is also varied among the runs, following its distribution in the SYM models.

Because the “open” areas are managed through days at sea, LPUE needs to be estimated in these areas. This is done using a linear regression between mean exploitable biomass in the open areas, and observed LPUE in these areas. The SAMS model gives projections of mean open area exploitable biomass, which then are translated into projected LPUE using the regression.

Figure 4.2 shows the one year projection errors made by the SAMS model in recent years. Mid-Atlantic projected biomass has been consistently overestimated. These projections assumed $M = 0.25$ for the Mid-Atlantic (except the two most southern areas). This is further evidence that M has been underestimated in this area. Additionally, recruitment in the Mid-Atlantic has been well below average. In future years, Mid-Atlantic recruitment will be reduced consistent with the SYM model. Georges Bank projections showed a much smaller overestimation. The increase in M from 0.2 to 0.27 will reduce this bias. Additionally, recruitment on Georges Bank during these years has been somewhat below average, although there is no evidence of an environmental shift as was observed in the Mid-Atlantic.

GeoSAMS

While SAMS was an advancement over non-spatial models, it is still spatially crude compared to the scale that the fishery operates. For example, SAMS assumes that effort and fishing mortality are spatially uniform inside each SAMS area, whereas in reality, fishing mortality can be highly heterogeneous because scallops are attracted to areas with highest catch rates (Hart 2001). In addition, it is difficult to reconfigure SAMS to a new spatial configuration; a more flexible approach was desired.

To resolve these issues, work began on a GeoSAMS, a more spatially explicit version of the SAMS model. In GeoSAMS, population dynamics occurs at each survey location in the initial year, using methods very similar to the SAMS model. After each model year, the estimates at the survey points are interpolated to a fine scale grid, using a regression

kriging methodology similar to that used for the Habcam survey. This allows for estimates of biomass and landings from any desired area.

Unfortunately, funding for the GeoSAMS project was halted in early 2025, when it was nearing but not quite operational status. Figure 4.3 shows results from a preliminary run of GeoSAMS, forecasting 2023 biomass from 2022 survey results.

Reference

Hart, DR (2001). Individual-based yield-per-recruit analysis, with an application to the Atlantic sea scallop, *Placopecten magellanicus*. Canadian Journal of Fisheries and Aquatic Sciences, 58(12):2351-2358.

Figures

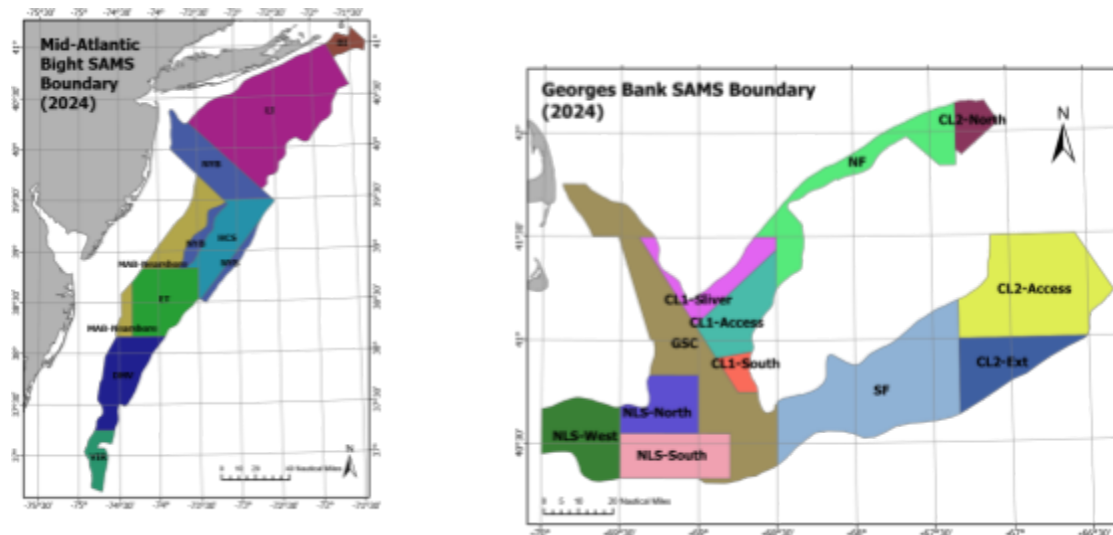


Figure 4.1. Mid-Atlantic and Georges Bank areas used in the SAMS projection model. The Gulf of Maine areas are not shown.

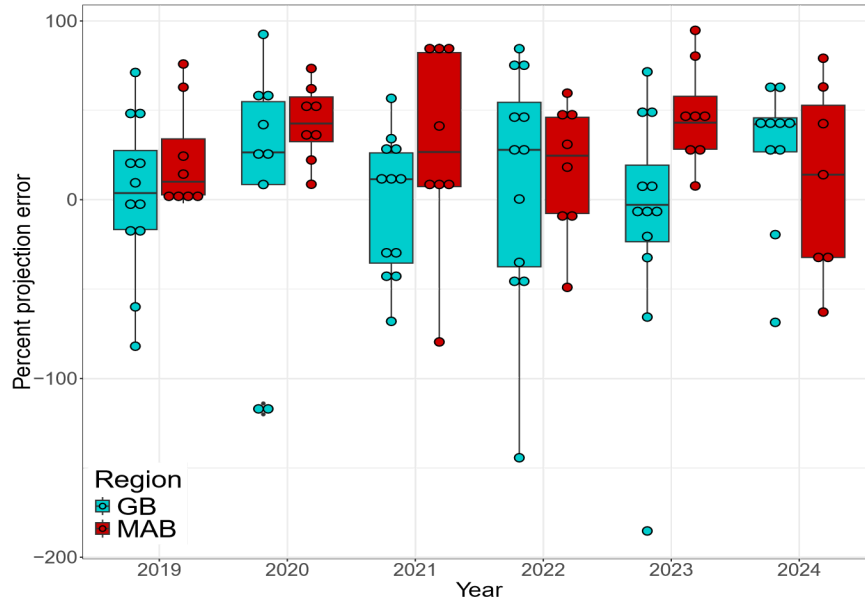


Figure 4.2. Realized biomass projection error by year, region and SAMS area. Projections were made in 2018-2023 to the years 2019-2024. A positive error indicates that SAMS overestimated biomass.

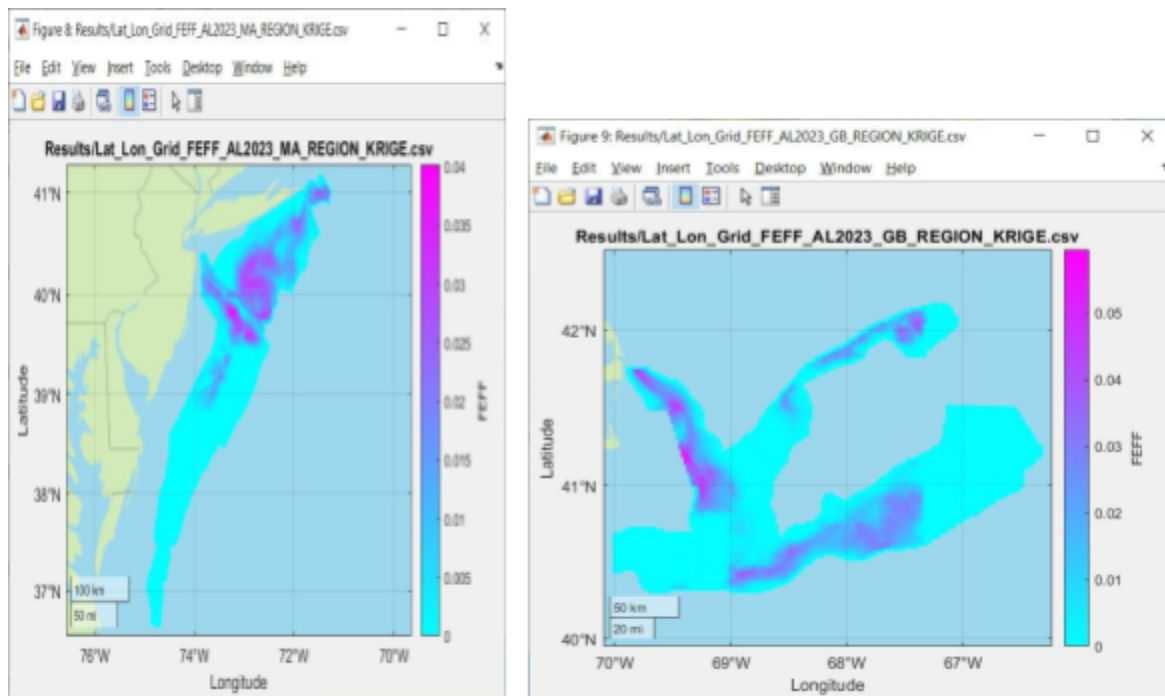


Figure 4.3. Example forecast from GeoSAMS from 2022 to 2023. Note that the areas surveyed by the CFF Habcam (Closed Area II South and Nantucket Lightship) were not included in the Georges Bank model.

Appendix – Technical Documentation for the SAMS model

Subareas were chosen to coincide with current management. In particular, Georges Bank was divided into three open areas (the Great South Channel, Northern Edge, Southern Flank), eight rotational or closed areas: Closed Areas I North and Middle, Closed Area II North and South, Closed Area II Extension, and three portions of the Nantucket Lightship Closed Area access area: northeast, west, and south deep (“Peter Pans”). The Mid-Atlantic was subdivided into seven areas: Virginia, Delmarva, Elephant Trunk, Hudson Canyon South, New York Bight, Nearshore New York Bight, and Long Island. Recently, the Gulf of Maine was included in the SAMS model. It is represented by five areas: Stellwagen South, Stellwagen NGOM, Ipswich Bay, Western Gulf of Maine Closure, and all other areas, including Jefferies Ledge.

The model tracks population vectors $\mathbf{p}(i,t) = (p_1, p_2, \dots, p_n)$, where $p_j(i,t)$ represents the density of scallops in the j th size class in area i at time t . The model uses a difference equation approach, where time is partitioned into discrete time steps $t_1, t_2, \dots$, with a time step of length $\Delta t = t_{k+1} - t_k$. The landings vector $\mathbf{h}(i,t_k)$ represents the catch at each size class in the i th region and k th time step. It is calculated as:

$$\mathbf{h}(i,t_k) = [I - \exp(\Delta t \mathbf{H}(i,t_k))] \mathbf{p}(i,t_k),$$

where I is the identity matrix and H is a diagonal matrix whose j th diagonal entry h_{jj} is given by:

$$h_{jj} = 1 / (1 + \exp(s_0 - s_1 * s))$$

where s is the shell height of the mid-point of the size-class.

The landings $L(i,t_k)$ for the i th region and k th time step are calculated using the dot product of landings vector $\mathbf{h}(i,t_k)$ with the vector $\mathbf{m}(i)$ representing the vector of meat weights at shell height for the i th region:

$$L(i,t_k) = A_i \mathbf{h}(i,t_k) \cdot \mathbf{m}(i)$$

where e_i represents the dredge efficiency in the i th region.

Even in the areas not under special area management, fishing mortalities tend not to be spatially uniform due to the sessile nature of sea scallops (Hart 2001). Fishing mortalities in open areas were determined by a simple “fleet dynamics model” that estimates fishing mortalities in open areas based on area-specific catch rates, and so that the overall DAS or open-area F matches the target. Based on these ideas, the fishing mortality F_i in the i th region is modeled as:

$$F_i = k * f_i * L_i$$

where L_i is the estimated LPUE (landings per day charged) in the i th region, f_i is an area-specific adjustment factor to take into account preferences for certain fishing grounds (due to lower costs, shorter steam times, ease of fishing, habitual preferences, etc.), and k is a

constant adjusted so that the total DAS or fishing mortality meets its target. For these simulations, $f_i = 1$ for all areas.

Scallops of shell height less than a minimum size s_d are assumed to be discarded, and suffer a discard mortality rate of d , taken here, as in the rest of the assessment to be 20%. There is also evidence that some scallops not actually landed may suffer mortality due to incidental damage from the dredge. Let F_L be the landed fishing mortality rate and F_I be the rate of incidental mortality on scallops not caught. For Georges Bank, which is a mix of sandy and hard bottom, we used $F_I = 0.2F_L$. For the Mid-Atlantic (almost all sand), we used $F_I = 0.1F_L$. Incidental mortality for a given shell height bin was then calculated using equations (4.3) and (4.4) of the main document.

Growth in each subarea was specified by a growth transition matrix G , based on area-specific growth increment data. Recruitment was modeled stochastically, and was assumed to be log-normal in each subarea. The mean, variance and covariance of the recruitment in a subarea was set to be equal to that observed in the historical time-series. New recruits enter the first size bin at each time step at a rate r_i depending on the subarea i , and stochastically on the year. Recruitment was modified by region-wide stock-recruit relationships based on those from the SYM model (TOR-5).

The population dynamics of the scallops in the present model can be summarized in the equation:

$$p(i, t_{k+1}) = \rho_i + G \exp(-M\Delta t H) p(i, t_k),$$

where ρ_i is a random variable representing recruitment in the i th area. The model was run with 10 time steps per year. The population and harvest vectors are converted into biomass by using the shell-height meat-weight relationship:

$$W = \exp[a + b \ln(s)],$$

where W is the meat weight of a scallop of shell height s . These relationships are subarea-specific. For calculating biomass, the shell height of a size class was taken as its midpoint.

LPUE is estimated using a linear regression relating exploitable biomass to LPUE. In the model, the regression is applied to projected exploitable biomass by area to predict LPUE